

Advanced Comment Analysis System

Connected Intelligence — Tanta University · Faculty of Commerce · Business Information Systems Program (BIS) · 2026

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The Inspiration: A Story of Being Unheard

We all know how frustrating it feels when you complain about something and **nobody responds**. You leave a review, send a message, post a comment — and nothing happens. That experience is more common than it should be, and it costs companies the trust of their customers every single day and the **most valuable asset company have are the loyal customer so all companies fight over them by keeping them satisfied**.

The truth is, it's not always because companies don't care. It's because the **volume of feedback they receive is simply too large to handle manually**. Comments pile up across social media, review sites, and complaint forms, and somewhere in that pile, real problems go unnoticed.

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THE SPARK

One Teammate's Breaking Point

This is exactly what happened to one of our teammates. He's a frequent traveler who keeps losing his luggage with the same airline. He complained. He posted. He wrote comment after comment. **Nothing changed. Nobody seemed to be listening.**

That experience frustrated him enough to actually do something about it. He built a tool that uses **AI and web scraping** to automatically collect and analyze customer comments — so airlines could finally catch the feedback they were missing.

When he showed us what he'd built, we saw an opportunity to take it further. Because this isn't just an airline problem. **Every company, in every industry, faces the same challenge.**

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Too Much Feedback, Not Enough Time

Every organization trying to maintain its customer base eventually hits the same wall. The challenge is universal — and the cost of ignoring it is real.

The Problem

Thousands of voices speaking at once through every channel — social media, review sites, complaint forms.

The Gap

Human teams are drowning, making it impossible to identify urgent issues before they escalate into PR nightmares.

The Cost

Every unaddressed comment risks a critical "blind spot" that compromises customer satisfaction and long-term brand loyalty.

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Introducing ACAS

Listen at Scale

So we expanded the idea and built **ACAS** — the **Advanced Comment Analysis System**. A smart platform that reads customer comments, understands the sentiment behind them, and helps companies take action before small problems become big ones.

From Airport Chaos to Universal Intelligence

ACAS began solving a very specific problem in the high-pressure world of aviation — massive volumes of passenger complaints about lost luggage and service bottlenecks. As the system matured, we realized the same logic applied to retail, healthcare, tech, and every sector in between.

The Pivot: Almost 100% Flexible Framework

We stripped away industry-specific constraints and rebuilt ACAS from the ground up — transforming it from an "airport tool" into a universal platform designed for any company, in any sector.

> *"We started by solving the chaos of the runway; we finished by creating a system that helps the whole world listen."*

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Goals & Operational Impact

ACAS eliminates feedback "blind spots" by transforming massive volumes of raw noise into clear, actionable intelligence in real-time. Its almost 100% flexible design allows any department to proactively maintain customer relationships and solve operational bottlenecks before they escalate.



Total Visibility

Eliminates "blind spots" by analyzing near 100% of feedback, no matter the volume.



Proactive Response

Identifies urgent issues and negative sentiment before they become crises.



Time Efficiency

Automates thousands of hours of manual reading, freeing teams for high-level tasks.



Universal Agility

almost 100% flexible design that plugs into any department — HR, Marketing, Ops, or Support.



Data-Led Strategy

Converts raw human emotions into clear, objective metrics for better decision-making.

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Competitor Analysis: Why ACAS?

The current market offers several robust solutions, each with distinct strengths. However, a fundamental gap remains: the market lacks a truly comprehensive, all-in-one content analysis solution with deep linguistic specialization for underserved regions — particularly the **Middle East and Arabic language**.

ACAS advantage: A unified platform designed for universal deployment, with Arabic specialization at its core.

Competitor	Strength	Limitation
Brandwatch	Unparalleled depth in AI-driven sentiment analysis; sweeps millions of comments	High cost and complexity; restricted to global corporations
Sprout Social & Hootsuite	Optimized for daily social media management; accessible interfaces	Basic analytical capabilities; confined to owned social channels
Chattermill	Focuses on operational improvements via complaint categorization	Lacks breadth for general marketing insights
MonkeyLearn	Highly flexible, no-code platform for custom text classification	No inherent data fetching capabilities
ACAS	**All-in-one platform with Arabic language specialization, smart auto-routing, and 100% flexibility**	**Purpose-built for underserved markets and universal deployment**

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Core System Capabilities

ACAS delivers a robust suite of capabilities designed to optimize the analysis and management of user content — from raw data ingestion to actionable intelligence.



Specialized AI & Linguistic Analysis

Advanced AI trained with specialized data to perform automatic emotion detection (joy, anger, frustration) and intent classification (complaints, praise, bug reports) with reliable confidence scores.



Data Acquisition & Workflow Automation

Web scraping technology for extracting comments from specified URLs, combined with Smart Routing to automatically direct content to the appropriate departments based on predefined rules.



Efficient Content Management

Bulk management operations (classification, prioritization, deletion) supported by a multi-parameter filtering system for real-time search across all collected content.



Standardized Reporting & Compliance

Predefined response templates, data export in standard formats (CSV, JSON), and a complete log of user activities and timestamps for full accountability.

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Development & Environment – Tools Used

A clear breakdown of the technical stack behind ACAS, organized by the core tools used to build, train, ingest, and present the system.

Development & Environment

Tools: VS Code, PowerShell

Used for: Day-to-day coding, debugging, script execution, and local workflow automation.

Why chosen: VS Code provides a fast, flexible editing experience, while PowerShell supports efficient command-line control and repeatable development tasks.

Backend Architecture

Tools: Python, FastAPI, SQLite

Used for: API development, business logic, and lightweight data storage for prototyping and structured persistence.

Why chosen: Python offers rapid development and a strong ecosystem, FastAPI enables high-performance APIs with clean validation, and SQLite keeps the system simple, portable, and easy to deploy.

Artificial Intelligence & Machine Learning

Tools: Hugging Face Transformers, Multiclass Emotion Classification, Supervised Learning

Used for: Sentiment and emotion analysis, model training, and text classification across feedback sources.

Why chosen: Transformers deliver strong language understanding, multiclass classification supports nuanced emotion detection, and supervised learning makes the system measurable, adaptable, and trainable on labeled feedback data.

Data Ingestion & UI

Tools: Web Scraping with BeautifulSoup & Requests, Chart.js / Plotly

Used for: Collecting structured data from web sources and turning results into interactive visualizations for users.

Why chosen: BeautifulSoup and Requests (libararies) make scraping reliable and efficient, while Chart.js and Plotly provide clear, interactive charts that make insights easy to understand at a glance.

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Recommendations & Future Steps

1 Human-in-the-Loop Validation

Keep human moderators for "Critical" tickets to periodically audit AI emotional detection accuracy.

2 Multilingual Expansion

Integrate translation layers or multilingual Transformer models (mBERT) to analyze feedback in Arabic, French, and Spanish.

3 Voice-to-Text Integration

Add a module to process audio feedback from customer service hotlines and kiosks, expanding the "Digital Ear" to spoken word.

4 Predictive Crisis Modeling

Implement ML to identify patterns before a crisis — alerting management to operational failures before they escalate.

5 Cloud Scalability

Migrate from local SQLite to cloud-based PostgreSQL or MongoDB, scaling ACAS from thousands to millions of comments globally.

> *"By integrating multilingual support and predictive modeling, ACAS will evolve from a reactive management tool into a proactive intelligence platform, capable of safeguarding an organization's reputation before issues even arise."*

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